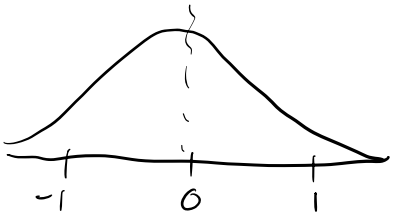
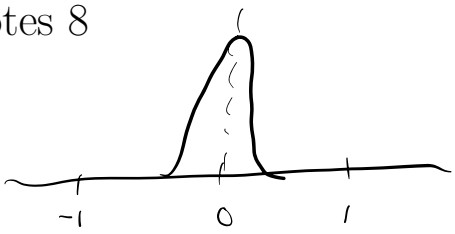
STAT 131: Lecture Notes 8



Winter 2024



8.1 Variance and Higher Moments [DS 4.3]

Previously we discussed how the mean, or expected value, can be used to summarize a distribution. The mean gives us an idea of what the average value for a random variable might be. Notice that it doesn't really tell us how far away from the mean a random variable might be. We will now introduce the concept of variance for this purpose.

Definition 8.1 (Variance)**.** The variance of a random variable X, written as $V(X)$ is defined as the expected squared difference of X from its mean:

$$V(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Note that we may also use $Var(X)$, σ_X^2 or σ^2 to denote the variance.

Example 8.1 (Variance of a Bernoulli)**.** Let $X \sim \text{Bernoulli}(p)$. Find $V(X)$.

$E[X] = 1 \cdot p + 0 \cdot (1-p) = p$ $\quad E[X^2] = 1^2 \cdot p + 0^2(1-p) = p$

$$V(X) = E[(X - E[X])^2] = E[(X-p)^2]$$

$$= E[X^2 - 2Xp + p^2] = E[X^2] - 2E[X]p + p^2$$

$$= p - 2 \cdot p \cdot p + p^2 = p - p^2 = p(1-p)$$

♢

Lemma 8.1 (Alternative expression of variance)**.** *The variance of a random variable X can equivalently be expressed as follows:*

$$V(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Proof. By definition.

$$V(X) = E\left[(X - E[X])^2\right] = E\left[X^2 - 2X \cdot E[X] + E[X]^2\right]$$

$$= E[X^2] - 2E[X] \cdot E[X] + \left(E[X]\right)^2$$

$$= E[X^2] - \left(E[X]\right)^2$$

□

Example 8.2 (Variance of a Bernoulli)**.** Let $X \sim \text{Bernoulli}(p)$. Find $V(X)$.

$$E[X] = p \qquad E[X^2] = p$$

$$V(X) = E[X^2] - (E[X])^2$$

$$= p - p^2 = p(1-p)$$

♢

Example 8.3 (Variance of a Uniform)**.** Let $X \sim \text{Uniform}(a, b)$. Find $V(X)$.

$$E[X] = \frac{a+b}{2} \quad , \quad E[X^2] = \frac{b^2 + ab + a^2}{3}$$

$$V(X) = E[X^2] - (E[X])^2 = \frac{b^2+ab+a^2}{3} - \frac{(a+b)^2}{4}$$

$$= \frac{4b^2+4ab+4a^2}{12} - \frac{3b^2+6ab+3a^2}{12} = \frac{b^2-2ab+a^2}{12} = \boxed{\frac{(b-a)^2}{12}}$$

♢

Definition 8.2 (Standard deviation)**.** The standard deviation of a random variable X is defined as

$$\sigma_x = \sqrt{V(X)}.$$

8.1.1 Properties of Variance

Elementary properties:

- $V(X) \geq 0$ for all X. $E[(X - E[X])^2]$
- $V(X) = 0$ iff $\mathbb{P}(X = c) = 1$ for some constant c. $Y = aX + b$

Theorem 8.2 (Variance of a linear function of X)**.** *Suppose X is a random variable, while a and b are two constants. Then*

$$V[aX + b] = a^2 V[X]$$

Example 8.4. Suppose you are playing a game where you flip a coin. If the coin lands on heads, you win \$5, but if the coin lands on tails, you win nothing. Let X represent the amount of money you receive from the game. Find $V(X)$.

Let $Z = \begin{cases} 1 & \text{if heads} \\ 0 & \text{if tails} \end{cases}$ Then $X = 5 \cdot Z$

$Z \sim \text{Bernoulli}(1/2) \Rightarrow V(Z) = 1/2(1 - 1/2) = 1/4$

$V(X) = V(5 \cdot Z) = 5^2 \cdot V(Z) = \boxed{25/4}$

♢

Theorem 8.3 (Linearity of Variance under Independence)**.** *Let X and Y be random variables where $X \perp Y$. Then,*

$$V(X + Y) = V(X) + V(Y).$$

This generalizes to show that if $X_1, X_2, \ldots, X_n$ are independent, then,

$$V(X_1 + X_2 + \cdots + X_n) = V(X_1) + V(X_2) + \cdots + V(X_n).$$

Proof. First statement only

$V(X+Y) = E[(X+Y)^2] - (E[X+Y])^2$

$= E[X^2 + 2XY + Y^2] - (E[X] + E[Y])^2$

$= E[X^2] + 2E[XY] + E[Y^2] - E[X]^2 - 2E[X]E[Y] - E[Y]^2$

$= (E[X^2] - E[X]^2) + (E[Y^2] - E[Y]^2) + 2E[XY] - 2E[X]E[Y]$

$= V(X) + V(Y) + 2E[XY] - 2E[X]E[Y]$

$= V(X) + V(Y)$ $\quad (\text{if } X \perp Y \Rightarrow E[XY] = E[X]E[Y])$

□

Example 8.5. Suppose that $X \sim \text{Binomial}(n, p)$. Find $\mathbb{E}[X^2]$.

$E[X^2] = \sum_x x^2 P(X=x) = \sum_{x=1}^{n} x^2 \binom{n}{x} p^x (1-p)^{n-x}$ ← hard to do!

$X = X_1 + X_2 + \cdots + X_n$ where $X_i \overset{ind}{\sim}$ Bernoulli(p)

So $V(X) = V(X_1) + V(X_2) + \cdots + V(X_n) = n \cdot V(X_i) = np(1-p)$

but $V(X) = E[X^2] - E[X]^2 = E[X^2] - (np)^2 = np(1-p)$

$\Rightarrow E[X^2] = np(1-p) + (np)^2$

♢

Example 8.6. Suppose that $X_1, X_2, \ldots, X_n$ are independent and identically distributed (i.i.d) $X_i \sim$ Exponential(λ). Find the variance of the sample mean, $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Hint: You may use that fact that the exponential distribution has variance $\frac{1}{\lambda^2}$.

$V(\bar{X}) = V\left(\frac{1}{n}[X_1 + X_2 + \cdots + X_n]\right) = \left(\frac{1}{n}\right)^2 V(X_1 + X_2 + \cdots + X_n)$

$= \left(\frac{1}{n}\right)^2 (V(X_1) + V(X_2) + \cdots + V(X_n)) = \left(\frac{1}{n}\right)^2 \cdot n\, V(X_i)$

$= \left(\frac{1}{n}\right)^2 \cdot n \cdot \frac{1}{\lambda^2} = \frac{1}{n\lambda^2}$ ← This goes to 0 as n becomes large!

♢

8.2 Covariance and Correlation [DS 4.6]

Suppose now that X and Y are two random variables that are not independent. Working with such random variables will require us to introduce the notion of covariance.

Definition 8.3 (Covariance)**.** The covariance of any two random variables X and Y, denoted by $Cov(X, Y)$, is defined as

$$Cov(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y]],$$

provided the expectation exists.

For two random variables, covariance is a numerical summary of their relationship—how do they vary together?

- If the larger than average values of X tend to happen with larger than average values of Y, then $Cov(X, Y)$ will be positive and we say X and Y are positively correlated.

- If the larger than average values of X tend to happen with smaller than average values of Y, then $Cov(X,Y)$ will be negative and we say X and Y are negatively correlated.

Example 8.7. Often times, people are interested in the relationship between two random variables.

- Verbal score and quant score of the GRE
- GRE score and the future academic GPA ←
- Number of classes missed and score on a final exam

♢

Lemma 8.4 (Alternative definition of covariance)**.** *Alternatively, we can write*

$$Cov(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Proof. Use the definition of covariance.

$$Cov(X,Y) = E[(X-E[X])(Y-E[X])]$$
$$= E[XY - XE[Y] - E[X]Y + E[X]E[Y]]$$
$$= E[XY] - E[X]E[Y] - E[X]E[Y] + E[X]E[Y]$$
$$= E[XY] - E[X]E[Y]$$

□

Example 8.8. Suppose the joint pdf of (X,Y) is

$$f(x,y) = 2xy + 1/2 \text{ for } 0 \le x \le 1 \text{ and } 0 \le y \le 1.$$

Show that $Cov(X,Y) = 1/144$.

$f_X(x) = \int_0^1 2xy + \frac{1}{2}\,dy = xy^2 + y/2\Big|_0^1 = x + 1/2 \Rightarrow E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x)\,dx = \int_0^1 x(x+\frac{1}{2})\,dx = \frac{x^3}{3} + \frac{x^2}{4}\Big|_0^1 = \frac{1}{3} + \frac{1}{4} = \boxed{7/12}$

Similarly, $E[Y] = 7/12$

$E[XY] = \int_0^1 \int_0^1 xy(2xy + \frac{1}{2})\,dx\,dy = \int_0^1 \int_0^1 2x^2y^2 + \frac{xy}{2}\,dx\,dy = \int_0^1 \left(\frac{2}{3}x^3y^2 + \frac{x^2y}{4}\Big|_0^1\right) dy$

$= \int_0^1 \frac{2y^2}{3} + \frac{y}{4}\,dy = \frac{2}{9}y^3 + \frac{1}{8}y^2\Big|_0^1 = \frac{2}{9} + \frac{1}{8} = \frac{25}{72} = \frac{50}{144}$

$\Rightarrow Cov(X,Y) = E[XY] - E[X]E[Y] = \frac{50}{144} - \left(\frac{7}{12}\right)^2 = \frac{50}{144} - \frac{49}{144} = \boxed{1/144}$

♢

Elementary properties of covariance

1. In contrast to variances which are non-negative, covariance can be negative.
2. $Cov(X, X) = V(X)$;
3. If $X \perp Y$, then $Cov(X, Y) = 0$.
4. It is also possible for $Cov(X, Y) = 0$ but X and Y are dependent.

Theorem 8.5 (Variance of a sum)**.** *Let X and Y be two random variables. Then,*

$$V(X + Y) = V(X) + V(Y) + 2Cov(X, Y).$$

Notice that this formula does not require any assumptions about the relationship between X and Y.

A key drawback of covariance is that it is not unit free. For example, the covariance between two random distance variables will be different depending on whether you measure distance in feet vs. inches. A solution is to rescale the covariance.

Definition 8.4 (Correlation)**.** The correlation between two random variables X and Y is defined as

$$corr(X, Y) = \frac{cov(X, Y)}{\sqrt{V(X)}\sqrt{V(Y)}},$$

Alternatively, we may write $\rho(X, Y) = \frac{cov(X,Y)}{\sigma_X \sigma_Y}$. Note that correlation will always be between -1 and 1.

8.3 Conditional Expectations [DS 4.7]

We are often interested in the expected value of a random variable conditioned on some event. For example, if X is the price of a hotel room on the beach, and A is the event that the current month is June, we might want to calculated $\mathbb{E}[X|A]$, or the expected value of the hotel room given that the month is June.

Definition 8.5. Given two random variables X and Y,

- If $Y|X = x$ is discrete, $\mathbb{E}[Y|X = x] = \sum y f_{Y|X}(y|x)$
- If $Y|X = x$ is continuous, $\mathbb{E}[Y|X = x] = \int y f_{Y|X}(y|x)dy$

$E[X] = \sum x f(x)$

$E[X] = \int x f(x) dx$

Example 8.9. Suppose the random variable X has the following pmf:

$$f_X(x) = \begin{cases} 0.1, \text{ for } x = 1 \\ 0.2, \text{ for } x = 2 \\ 0.3, \text{ for } x = 3 \\ 0.2, \text{ for } x = 4 \\ 0.2, \text{ for } x = 5 \end{cases} \quad \begin{matrix} \left.\begin{matrix} \\ \\ \\ \end{matrix}\right] Y=1 \\ \left.\begin{matrix} \\ \end{matrix}\right] Y=0 \end{matrix}$$

Now suppose that Y is a random variable that is equal to 1 if $X \leq 3$ and 0 otherwise. What is $\mathbb{E}[X]$? What is $\mathbb{E}[X|Y=1]$?

$E[X] = \sum_x x f(x) = 1 \cdot 0.1 + 2 \cdot 0.2 \cdots 5 \cdot (0.2)$

$$E[X|Y=1] = \sum_x x \cdot f_{X|Y}(x|y=1) = \sum_x x \frac{f(x, Y=1)}{f_Y(Y=1)}$$

$$= 1 \cdot \frac{0.1}{0.6} + 2 \cdot \frac{0.2}{0.6} + 3 \cdot \frac{0.3}{0.6}$$

$$= 1 \cdot \frac{1}{6} + 2 \cdot \frac{2}{6} + 3 \cdot \frac{3}{6} = \boxed{14/6}$$

♢

Example 8.10. Suppose that two random variables X and Y have the following joint pdf:

$$f(x, y) = x + y \text{ for } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1.$$

Compute $\mathbb{E}[Y|X = \frac{1}{2}]$.

$$f_X(x) = \int_0^1 x + y \, dy = xy + \frac{y^2}{2} \Big|_0^1 = x + 1/2 \qquad f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{x+y}{x+1/2}$$

$$\Rightarrow f_{Y|X}(y|x=1/2) = \frac{1/2 + y}{1/2 + 1/2} = y + 1/2$$

$$\Rightarrow E[Y|X=1/2] = \int_0^1 y \, f_{Y|X}(y|x=1/2) = \int_0^1 y(y+1/2) \, dy$$

$$= \frac{y^3}{3} + \frac{y^2}{4} \Big|_0^1 = \boxed{7/12}$$

♢

Consider that if we know the value of $X = x$, then $\mathbb{E}[Y|X = x]$ is just some constant value that we can calculate. However, if we don't know the value of X, then $\mathbb{E}[Y|X]$ is also a random variable, and a function of X (it will change for different values of X). Since $\mathbb{E}[Y|X]$ is itself a random variable, we can find its expected value by integrating/summing over all possible values of X. This leads to the law of iterated expectation.

Theorem 8.6 (Law of Iterated Expectation (LIE))**.**

$$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y].$$

Proof. Continuous case.

$$E[E[Y|X]] = \int_{-\infty}^{\infty} E[Y|X] \cdot f_X(x)\,dx$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} y\, f_{Y|X}(y|x) f_X(x)\,dy\,dx$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} y\, f(x,y)\,dy\,dx = E[Y]$$

□

Example 8.11. Suppose n patients are recruited for a clinical trial of a new medication. Let X be the number of patients that had successful treatment. Also suppose that the probability of success p follows a uniform distribution (i.e., $p \sim \text{Uniform}(0,1)$). What is $\mathbb{E}[X]$?

$$[X|p] \sim \text{Binomial}(n,p) \quad , \quad p \sim \text{Uniform}(0,1)$$

$$E_X[X|p] = np$$

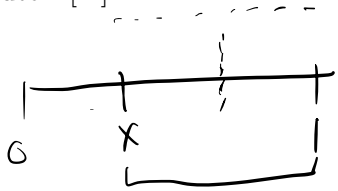
$$E[E[X|p]] = E_p[np] = nE[p] = n \cdot \tfrac{1}{2} = \boxed{n/2}$$

♢

Example 8.12. Suppose $X \sim Unif(0,1)$, $[Y|X=x] \sim Unif(x,1)$, compute $\mathbb{E}[Y]$.

$$E[E[Y|X]] = E[Y]$$

$$E[Y|X] = \frac{X+1}{2}$$



$$E\left[\frac{X+1}{2}\right] = \frac{1}{2} + \frac{1}{2}E[X] = \frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2} = \boxed{3/4}$$

♢

8.4 Conditional Variance [DS 4.7]

Definition 8.6. For any given value of x, let $V(Y|X=x)$ denote the variance of the conditional distribution of Y given $X=x$. In other words,

$$V(Y|X=x) = \mathbb{E}[(Y - \mathbb{E}(Y|X=x))^2|X=x] = \mathbb{E}[Y^2|X=x] - (\mathbb{E}[Y|X=x])^2.$$

We call this the conditional variance of Y given $X=x$.

$$V(Y) = E[(Y - E(Y))^2]$$

Example 8.13. Suppose the random variables X and Y have the following joint pdf:

$$f(x,y) = x + y \text{ for } 0 \le x \le 1 \text{ and } 0 \le y \le 1.$$

Find $V(Y|X)$.

$$f_{Y|X}(y|x) = \frac{x+y}{x+1/2}$$

$$E[Y|X] = \int_0^1 y \cdot \frac{x+y}{x+1/2}\, dy = \frac{x/2 + 1/3}{x+1/2}$$

$$E[Y^2|X] = \int_0^1 y^2 \cdot \frac{x+y}{x+1/2}\, dy = \frac{x/3 + 1/4}{x+1/2}$$

$$V(Y|X) = E[Y^2|X] - (E[Y|X])^2 = \left(\frac{x/3 + 1/4}{x+1/2}\right) - \left(\frac{x/2+1/3}{x+1/2}\right)^2$$

♢

Theorem 8.7 (Law of total variance)**.** *Eve's law:*

$$V(Y) = E[V(Y|X)] + V[E(Y|X)].$$

Example 8.14. Suppose that the number of customers that shop at a store on any given day is denoted as N, where $\mathbb{E}[N] = 10$ and $V(N) = 10$. Let X_i represent the amount of money that the ith customer spends at the store. Also assume that the customers shop independently with $\mathbb{E}[X_i] = 5$ and $V(X_i) = 2$. Finally let $Y = \sum_{i=1}^{N} X_i$ represent the stores total sales. Find $\mathbb{E}[Y]$ and $V(Y)$.

$E[Y|N] = \sum_{i=1}^{N} E[X_i] = 5N \qquad V(Y|N) = \sum_{i=1}^{N} V(X_i) = 2 \cdot N$

$E[V(Y|N)] = E[2N] = 2 \cdot E[N] = 2 \cdot 10 = 20$

$V[E[Y|N]] = V[5 \cdot N] = 25 \cdot V[N] = 25 \cdot 10 = 250$

$\Rightarrow V(Y) = E[V(Y|N)] + V(E[Y|N]) = 20 + 250 = 270$

♢

8.5 Moment Generating Functions [DS 4.4]

Definition 8.7 (Moments)**.** $\mathbb{E}[X^k]$ for $k = 1, 2, ...$

Definition 8.8 (Central moments)**.** $\mathbb{E}[(X - \mathbb{E}(X))^k]$ for $k = 2, 3, ...$

- The mean or expectation of X is $\mathbb{E}[X] = \mathbb{E}[X^k]$ with $k = 1$. In other words, the mean is the first moment.
- The variance of X is $V(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[(X - \mathbb{E}[X])^k]$ with $k = 2$, so the variance is the second central moment.
- We can also find the variance as $V(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, using the first two moments.

We may want to summarize all the information about moments with one single function.

Definition 8.9. The moment generating function (mgf) of a random variable X is defined as

$$\psi_X(t) = \mathbb{E}(e^{tX})$$

for any real number t.

We see immediately that $\psi_X(0) = 1$ for all random variable X.

As the moment generating function is the expectation of a function of X, it is possible that the mgf does not exist for certain random variables. It turns out the moment generating function is a defining feature of random variables, provided its existence. In other words, if two random variables have the same moment generating function, then they both follow the same distribution.

Example 8.15. Suppose $X \sim \text{Bernoulli}(p)$. Find the mgf of X.

$$\psi_X(t) = E[e^{tX}] = \sum_{x=0}^{1} e^{tx} \cdot f(x)$$

$$= e^{t0} \cdot (1-p) + e^{t \cdot 1} p$$

$$= (1-p) + p e^t$$

♢

Example 8.16. Suppose $X \sim \text{Exponential}(\lambda)$. Find the mgf of X.

$$\psi_X(t) = E[e^{tx}] = \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx$$

$$= \lambda \int_0^\infty e^{(t-\lambda)x} dx = \lambda \left[\frac{1}{t-\lambda} e^{(t-\lambda)x} \Big|_0^\infty \right] = \lambda \left[0 - \frac{1}{t-\lambda} \right]$$

$$= \frac{\lambda}{\lambda - t} \quad \text{as long as } \lambda > t$$

♢

Theorem 8.8. *If $\psi(t) < \infty$ for all values of t in some open interval around $t = 0$, then*

$$\mathbb{E}[X^k] < \infty,$$

and, for $k = 1, 2, ...,$

$$\mathbb{E}[X^k] = \psi^{(k)}(0).$$

In other words, $\mathbb{E}[X^k]$ is equal to the kth derivative of $\psi(t)$ evaluated at the point $t = 0$.

Example 8.17. Suppose $X \sim \text{Bernoulli}(p)$. Find $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.

$$\frac{d}{dt}\psi_X(t) = \frac{d}{dt}(1-p) + pe^t = pe^t \qquad E[X] = pe^t\Big|^{t=0} = p$$

$$\frac{d^2}{dt^2}\psi_X(t) = \frac{d}{dt}pe^t = pe^t \Rightarrow E[X^2] = pe^t\Big|^{t=0} = p$$

♢

Example 8.18. Suppose $X \sim \text{Exponential}(\lambda)$. Find $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.

$$\frac{d}{dt}\psi_X(t) = \frac{d}{dt}\frac{\lambda}{\lambda - t} = \frac{\lambda}{(\lambda-t)^2} \Rightarrow E[X] = \frac{\lambda}{(\lambda-t)^2}\Big|^{t=0} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$$

$$\frac{d^2}{dt^2}\psi_X(t) = \frac{d}{dt}\frac{\lambda}{(\lambda-t)^2} = \frac{2\lambda(\lambda-t)}{(\lambda-t)^4} \Rightarrow E[X^2] = \frac{2\lambda(\lambda-t)}{(\lambda-t)^4}\Big|^{t=0} = \frac{2\lambda^2}{\lambda^4} = \frac{2}{\lambda^2}$$

♢

8.6 Properties of moment generating functions

Theorem 8.9 (Linear Transformation)**.** *Let X be a random variable with mgf $\psi_X(t)$ and $Y = aX + b$ for constants a and b. Then,*

$$\psi_Y(t) = e^{bt}\psi_X(at)$$

for all values of t such that $\psi_X(at)$ is finite.

Example 8.19. Suppose $X \sim \text{Exponential}(1)$ and $Y = 3 - 2X$. Find the mgf of Y.

$$\psi_X(t) = \frac{1}{1-t} \qquad a = -2 \qquad b = 3$$

$$\psi_Y(t) = e^{bt}\psi_X(at)$$

$$= e^{3t}\frac{1}{1+2t}$$

♢

Theorem 8.10 (MGF of Independent Sums)**.** *Let $X_1, \ldots, X_n$ be independent random variables, each with mgf $\psi_{X_i}(t)$ for $i = 1, \ldots, n$. Also let $Y = X_1 + \cdots + X_n$. Then,*

$$\psi_Y(t) = \prod_{i=1}^{n} \psi_{X_i}(t).$$

Example 8.20. Suppose $X \sim \text{Binomial}(n, p)$. Find the mgf of X.

$X = X_1 + X_2 + \cdots + X_n$ where $X_i \overset{iid}{\sim} \text{Bernoulli}(p)$

$$\psi_Y(t) = \prod_{i=1}^{n} \psi_{X_i}(t)$$

$$= \prod_{i=1}^{n} \left[(1-p) + p e^t \right]$$

$$= \left(1 - p + p e^t \right)^n$$

♢